

33. A system is excited by a signal,

$$x(t) = 4 \operatorname{rect}(t/2)$$

and its response is

$$y(t) = 10 \left[\left(1 - e^{-(t+1)} \right) u(t+1) - \left(1 - e^{-(t-1)} \right) u(t-1) \right]$$

What is its impulse response?

34. Graph the magnitudes and phases of the CTFTs of the following functions.

- (a) $g(t) = 5\delta(4t)$
- (b) $g(t) = 4[\delta_4(t+1) - \delta_4(t-3)]$
- (c) $g(t) = u(2t) + u(t-1)$

39. Answer the following questions.

- (a) A signal $x_1(t)$ has a CTFT $X_1(f)$. If $x_2(t) = x_1(t+4)$, what is the relationship between $|X_1(f)|$ and $|X_2(f)|$?
- (b) A signal $x_1(t)$ has a CTFT, $X_1(f)$. If $x_2(t) = x_1(t/5)$, what is the relationship between the maximum value of $|X_1(f)|$ and the maximum value of $|X_2(f)|$?
- (c) A CTFT has the value $e^{-j\pi/4}$ at a frequency $f = 20$. What is the value of that same CTFT at a frequency of $f = -20$?

47. Find the CTFT (either form) of the signal in figure E.47 and graph its magnitude and phase versus frequency on separate graphs. (Let $A = -B = 1$ and let $t_1 = 1$ and $t_2 = 2$.) Hint: Express this signal as the sum of two functions and use the linearity property.

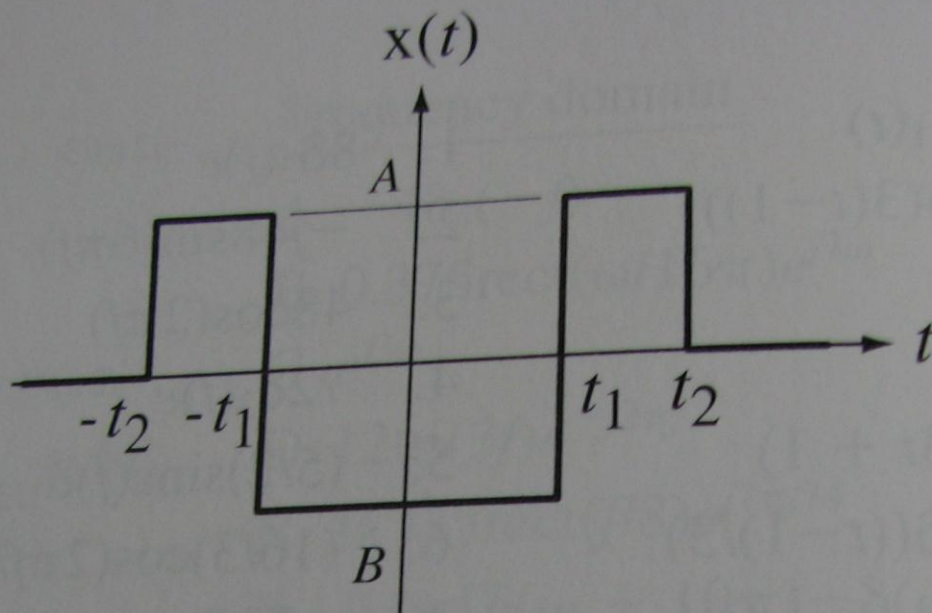


Figure E.47
A function.

48. In many communication systems, a device called a **mixer** is used. In its simplest form, a mixer is simply an analog multiplier. That is, its output signal $y(t)$ is the product of its two input signals. If the two input signals are

$$x_1(t) = 10 \operatorname{sinc}(20t) \quad \text{and} \quad x_2(t) = 5 \cos(2000\pi t)$$

graph the magnitude of the CTFT of $y(t)$, which is $Y(f)$, and compare it to the magnitude of the CTFT of $x_1(t)$. In simple terms, what does a mixer do?

49. One major problem in real instrumentation systems is electromagnetic interference caused by the 60 Hz power lines. A system with an impulse response of the form $h(t) = A(u(t) - u(t - t_0))$ can reject 60 Hz and all its harmonics. Find the numerical value of t_0 that makes this happen.